



Quantum Bluff

FINAL PROJECT REPORT

University of Strasbourg

Bachelor of Science in Computer Science (L3)

Supervised project — reference specification: Docs/Cahier_de_charge_6A_PI.pdf

Academic supervisor: *[to be completed by the team]*

Project manager: Omar

Team members: Linda, Massi, Abdoulaye, Soheil, Mohamed, Azra, Yigit

Academic year 2025–2026

Document generated: 14 May 2026

Abstract

Quantum Bluff is a capstone web platform produced for the third-year Bachelor's degree in Computer Science (L3) at the University of Strasbourg. It combines an authoritative Texas Hold'em stack with extended casino-style modules (solo and multiplayer blackjack, slots, European roulette), structured tournaments with ancillary winner markets, social features (friends, invitations, peer loans), and progression systems (XP, daily challenges, leaderboards). This report explains how delivery evolved from an initial V-model framing to an agile-inspired workflow; which engineering tools preserved modularity and testability; and how runtime architecture separates the React client, Express REST endpoints, a Socket.IO gateway, Prisma-backed persistence, and Redis-backed coordination. Security controls (JWT and optional TOTP, segmented rate limits, idempotency on sensitive mutations, anti-cheat hooks) and an automated test harness are summarized alongside appendices that map each major domain to repository artefacts. The product uses **virtual chips** for pedagogical demonstration; it does not implement real-money gambling.

Résumé

Quantum Bluff est une plateforme web temps réel développée dans le cadre du projet de fin d'études de L3 informatique à l'Université de Strasbourg. Elle couvre le poker Texas Hold'em (tables cash, bots, paris latents), des modules type casino (blackjack solo et multijoueur, machine à sous, roulette européenne), des tournois structurés avec marchés annexes sur le vainqueur, des fonctions sociales (amis, invitations, prêts entre joueurs) et de la progression (XP, défis quotidiens, classements). Ce rapport présente l'évolution de la gestion de projet (phase en V initiale puis itérations inspirées du Scrum), les choix technologiques, l'architecture logicielle, la sécurité et les résultats des tests automatisés. Des annexes indexent les chemins du dépôt pour chaque sous-système. Le système repose sur une **monnaie virtuelle** à visée pédagogique et ne constitue pas un service de jeu d'argent réel.

Keywords: multiplayer games, Texas Hold'em, real-time web, WebSocket, TypeScript, React, Node.js, Express, Prisma, PostgreSQL, Redis, Socket.IO, software engineering, security, automated testing.

Mots-clés : jeux multijoueurs, Texas Hold'em, web temps réel, WebSocket, TypeScript, React, Node.js, Express, Prisma, PostgreSQL, Redis, Socket.IO, génie logiciel, sécurité, tests automatisés.

List of abbreviations

API	Application Programming Interface (HTTP/JSON in this project).
CI/CD	Continuous integration and continuous deployment (GitLab CI at repository root).
HUD	Heads-up display: in-table probability and hand-category feedback.
JWT	JSON Web Token used for HTTP and Socket.IO authentication.
ORM	Object–relational mapping (Prisma).
REST	Representational state transfer style for <code>/api/*</code> routes.
RTK	Redux Toolkit and RTK Query for client-side data fetching.
TOTP	Time-based one-time password (optional second factor).
TTL	Time-to-live for cached or in-memory sessions.
UI/UX	User interface and user experience layers in the React SPA.
2FA	Two-factor authentication (TOTP in production paths).

Table of contents

Abstract	i
Résumé (français)	i
List of abbreviations	ii
1 Introduction	1
1.1 Context and Objectives	1
1.2 Scope of this document	1
2 Methodology, traceability, and reading guide	2
2.1 Traceability to the initial specification	2
2.2 Evidence hierarchy	2
2.3 How validation is reported	2
3 Project Evolution	3
3.1 Initial specifications (poker-first)	3
3.2 Scope expansion (casino, tournaments, social, progression)	3
3.3 Why the narrative moved from “catalog” to “appendices”	3
4 Project Management	4
4.1 Initial V-model phase	4
4.2 Transition to an agile-inspired workflow (Jira, Scrum practices)	4
4.3 Planning, coordination, and reporting	4
4.4 Product management mechanics	4
5 Development Tools and Technologies	5
5.1 Frontend and backend	5
5.2 Database and infrastructure	5
5.3 Version control and collaboration	5
5.4 Testing and deployment	5
6 Use of Artificial Intelligence	6
6.1 In-product AI: poker bots	6
6.2 AI-assisted engineering (tooling)	6
6.3 Limits and verification	6
7 Platform Functionalities	7
8 System Architecture	9
8.1 Monorepo layout	9
8.2 Real-time communication	9
8.3 Persistence and integrity	9

8.4	Data model overview (Prisma)	9
8.5	Diagram	9
9	Deployment and operational runbook	11
9.1	Local development stack	11
9.2	Production and desktop delivery	11
10	Security, Testing and Stability	12
10.1	Authentication and abuse controls	12
10.2	Testing and debugging	12
10.3	Automated test inventory and CI (reference snapshot)	12
10.4	Performance and stability	12
11	Technical Challenges	13
12	Ethics, privacy, and academic integrity	14
12.1	Virtual economy and regulatory framing	14
12.2	Personal data	14
12.3	Integrity of authorship and AI assistance	14
13	Individual Contributions	15
13.1	Omar – project management and architecture alignment	15
13.2	Other contributors	15
14	Limitations and Future Work	16
14.1	Current limitations	16
14.2	Future improvements	16
15	Conclusion	17
	Acknowledgements	18
A	Repository map and routing	19
A.1	Programmed routing and shell	19
B	Poker engine and cash game	20
B.1	What was programmed	20
B.2	Rules enforced	20
B.3	Further reading	20
C	Bot AI and practice integration	21
C.1	Programmed pipeline	21
C.2	Strength signal and decisions	21
C.3	Further reading	21

D	Probabilities, Quantum HUD, and Monte Carlo helper	22
D.1	Division of responsibilities	22
D.2	Monte Carlo equity estimator	22
D.3	Monte Carlo error budget (HUD analogue to RTP)	22
D.4	Quantum Combi—exact enumeration formula	23
D.5	Quantum Combi in the hand-ranking ladder (server and HUD)	23
E	Hidden betting system	24
E.1	Programmed architecture	24
E.2	Odds and tables (example)	24
E.3	Further reading	24
F	Tournaments and winner side bets	25
F.1	Programmed modules	25
F.2	Further reading	25
G	Blackjack solo and multiplayer	26
G.1	Solo blackjack engine	26
G.2	Further reading	26
H	Slot machine, roulette, and casino wallet	27
H.1	Three-reel slot model	27
H.2	Complete slot RTP (closed form, current constants)	27
H.3	European roulette model	28
H.4	Complete roulette RTP (all proportional inside bets)	28
H.5	Wallet and ledger	28
I	Social graph, invitations, and peer loans	29
I.1	Programmed surfaces	29
I.2	Further reading	29
J	Progression, challenges, and leaderboards	30
J.1	Programmed logic	30
J.2	Further reading	30
K	Authentication, profiles, and i18n	31
K.1	Programmed security and profiles	31
L	Operations, Redis, CI, and packaging	32
L.1	Programmed operations	32
M	External references and compendium	33

List of Figures

- | | | |
|---|--|----|
| 1 | High-level runtime flow (placeholder box; replace with exported diagram if desired). | 10 |
|---|--|----|

List of Tables

1	High-level functional inventory (representative surfaces).	7
2	Automated test inventory (representative snapshot; see <code>Docs/RAPPORT_TESTS.md</code>).	12
3	Individual contributions (template — team must fill before submission).	15

1 Introduction

1.1 Context and Objectives

Quantum Bluff was developed as part of the final third-year Bachelor’s degree project (L3 Computer Science, French academic system) at the University of Strasbourg. The objective was to deliver a **real-time multiplayer gaming platform** with a professional-grade engineering posture: modular architecture, automated testing, observability hooks, security hardening, and collaborative delivery under time constraints.

The **initial product intent** (before scope expansion) was a Texas Hold’em poker platform: practice mode against AI with multiple difficulty levels, multiplayer cash tables (public/private), configurable blinds, minimum buy-in rules, a probability-oriented HUD, and a **hidden side-bet** subsystem. The technical goals included authoritative server-side rules, transactional chip movements, WebSocket-scale event delivery, and operational readiness (health checks, metrics, recovery paths).

Security and reliability were first-class constraints from the outset: JWT authentication, optional TOTP 2FA, anti-cheat and rate-limiting layers, idempotent handling for sensitive mutations, and schema validation on critical API paths.

1.2 Scope of this document

This report summarizes **management evolution, tooling, delivered capabilities, and architecture**. Exhaustive technical narratives (bots, probabilities, tournaments, hidden bets, etc.) are deferred to **Appendices A–M** to avoid repetition. Repository-relative file paths use `\path{...}` notation. The front matter includes an abstract (English and French), a list of abbreviations, the table of contents, and lists of figures and tables.

2 Methodology, traceability, and reading guide

2.1 Traceability to the initial specification

The baseline expectations for the Bachelor project are captured in the repository specification `Docs/Cahier_de_charge_6A_PI.pdf`. The delivered system **extends** that poker-centric charter: all original poker-facing commitments (cash tables, bots, HUD, hidden bets, real-time play) remain grounded in server authority, while additional modules (casino mini-games, tournaments, social lending, progression) were integrated under the same engineering constraints (tests, rate limits, transactional chip movements). Where the PDF and the code diverge, **the implementation and this report defer to the repository as the operational source of truth**, with internal notes under `Docs/detail_logique/` providing rationale.

2.2 Evidence hierarchy

Claims in the body are intentionally **outcome-oriented**. File paths anchor verifiable detail. Long-form auto-generated material (for example `Docs/Rapport_final/RAPPORT_TECHNIQUE_COMPLET.md`) is treated as an *index*: useful for breadth, but not cited as automatically validated against every line of code.

2.3 How validation is reported

Automated suites and manual smoke scenarios are summarized in Section 10 and cross-checked against `Docs/RAPPORT_TESTS.md` and `Docs/rapport-validation.md`. Playwright end-to-end tests live under `client/e2e/`. CI behaviour is defined in `.gitlab-ci.yml`.

3 Project Evolution

3.1 Initial specifications (poker-first)

The early scope targeted: Texas Hold'em cash games; **AI practice** with four difficulty tiers (easy, medium, hard, expert) enforced in `server/src/routes/game.api.routes.ts` and `server/src/routes/bot.routes.ts`; public/private multiplayer rooms with SB/BB configuration and minimum balance rules; **real-time** updates via Socket.IO; a **Quantum HUD** for win-equity and hand-category feedback (`client/src/components/QuantumHUD.tsx`); and **hidden bets** resolved server-side (`server/src/poker/hiddenBets/`).

3.2 Scope expansion (casino, tournaments, social, progression)

The delivered system adds: **Blackjack** solo and **multi-table** lobbies (`client/src/pages/Blackjack.tsx`, `client/src/pages/BlackjackMultiLobby.tsx`, `client/src/pages/BlackjackMultiTable.tsx`); **slot** and **roulette** flows reachable from the mini-games hub (`client/src/pages/MiniGames.tsx`); **tournaments** with waiting room, in-tournament table, results, and **winner prediction side bets** (`client/src/features/tournament/`); **leaderboards**; **XP**, badges, daily challenges, daily login streak; **friends**, invitations, join requests, **peer loans** with interest/repayment hooks; **profile** editing with server-stored avatars; **2FA TOTP**; **feedback** and **player reports**; **wallet ledger**, gift codes, and dev recharge helpers; **i18n** in five locales (`client/src/i18n/locales/`); **tutorial** flow (`client/src/pages/TutorialGame.tsx`, `client/src/features/tutorial/`); **admin web console** and packaged targets (**Electron**, **Capacitor**) per `client/package.json`.

3.3 Why the narrative moved from “catalog” to “appendices”

The codebase grew across **client**, **server**, **database**, and **docs/** technical notes. Listing every subsystem in prose would duplicate `docs/ARCHITECTURE.md` and the structured logic notes under `Docs/detail_logique/`. The body therefore stays outcome-oriented; appendices map each major subsystem to code anchors.

4 Project Management

4.1 Initial V-model phase

The project began with a **V-model style** framing: business requirement gathering with end users in mind, benchmarking comparable products, brainstorming sessions **validated collectively** by the team, and a written requirements package plus an initial Gantt chart. A **technical lead** was appointed to structure the MOE (implementation) track while the project manager covered the MOA (needs, prioritization, specification) track until the specification baseline stabilized.

4.2 Transition to an agile-inspired workflow (Jira, Scrum practices)

After an intensive early phase (approximately **two months** of V-model-oriented execution on a **four-month** calendar), the team concluded that strict phase separation slowed integration of new gameplay and infrastructure ideas. The organization shifted toward an **iterative, Scrum-inspired workflow**: **Jira** for epics, user stories, and time-boxed sprints; backlog ordering by **value vs. risk**; twice-weekly synchronization meetings for decisions, unblockers, and demo-ready increments. The project manager studied agile practice material (including *Scrum Life*) to align rituals with the team's reality rather than copying a textbook process wholesale.

4.3 Planning, coordination, and reporting

Coordination relied on: recurring meetings (~twice per week), shared technical documentation under Docs/, internal validation sessions, and continuous integration feedback via `.gitlab-ci.yml` at the repository root. Work was distributed dynamically across frontend, backend, and infrastructure tasks because multiplayer features require tight coupling between UI state, socket events, and persistence.

4.4 Product management mechanics

Backlog prioritization balanced user-visible value (new game modes, social features) against enabling work (Redis adapter, rate limits, recovery jobs, observability). Cross-cutting concerns (security, idempotency, anti-cheat) were escalated early to avoid late integration risk.

5 Development Tools and Technologies

5.1 Frontend and backend

Client: React 19, TypeScript, Vite 7, Tailwind CSS, Radix UI primitives (`client/src/components/ui/`), react-router-dom v7, Redux Toolkit with RTK Query (`client/src/services/api.ts`), socket.io-client (`client/src/services/socket.ts`), i18next (`client/src/i18n/config.ts`).

Server: Node.js (ESM), Express 4, Socket.IO 4, Prisma 6 with PostgreSQL (`server/prisma/schema.prisma`), Redis clients for rate limiting, action logging, and the Socket.IO Redis adapter (`server/src/index.ts`), Zod validation, JWT auth, bcrypt password hashing, sanitize-html where applicable.

5.2 Database and infrastructure

Local PostgreSQL via `database/docker-compose.yml`. Redis for multi-instance coordination and abuse controls. Optional OpenTelemetry and Prometheus-oriented metrics endpoints (`server/src/observability/`).

5.3 Version control and collaboration

Git repository with GitLab CI configuration at `.gitlab-ci.yml`. Documentation and logic notes live under `Docs/` (including `docs/ARCHITECTURE.md` and `Docs/detail_logique/`).

5.4 Testing and deployment

Client: Vitest, Playwright (`client/e2e/`). Server: Jest and Supertest (`server/src/__tests__/_/`). Desktop/mobile packaging: Electron and Capacitor scripts in `client/package.json`.

6 Use of Artificial Intelligence

6.1 In-product AI: poker bots

The shipped product includes **heuristic poker bots** with configurable difficulty. Core decision logic lives in `server/src/logic/botAI.ts`, orchestrated HTTP services in `server/src/services/botAi.service.ts`, and practice-mode integration (`server/src/poker/services/practiceBotTurns.service.ts`). Difficulty is part of the validated API contract (`server/src/routes/bot.routes.ts`).

6.2 AI-assisted engineering (tooling)

Team members used modern **coding assistants** (e.g., IDE-integrated LLMs) for drafting, refactors, and documentation. Generated output was treated as **untrusted**: reviewed in merge requests, validated by automated tests, and rejected when it contradicted server authority or security invariants.

6.3 Limits and verification

Bots are **not** opaque learned models in production: they are inspectable code paths. Assistant-generated snippets were never merged without human review and test signal.

7 Platform Functionalities

Table 1 lists major domains. Details appear only once, in the referenced appendix, to prevent duplication. **Appendices A–M** also record implemented formulas: closed-form **slot RTP** and unified **roulette RTP**, Monte Carlo equity with an error bound for the **Quantum HUD**, **Quantum Combi** exact probability plus its place in the hand-ranking cascade, and the source files where each is coded.

Table 1: High-level functional inventory (representative surfaces).

Domain	User-visible capability	Primary UI entry	Deep dive
Poker cash	Multiplayer tables, pot logic, chat, actions	<code>/game,</code> <code>/waiting-room</code>	Appendix B
Poker bots	Practice tables, four difficulties	<code>/bot-configuration</code>	Appendix C
HUD	Win chance and hand category estimates	In-game HUD	Appendix D
Hidden bets	Side markets, pricing, settlement	In-game panel, <code>/results</code>	Appendix E
Blackjack	Solo + multi lobby/table	<code>/blackjack,</code> <code>/blackjack/</code> lobby, <code>/blackjack/</code> <code>table/:gameId</code>	Appendix G
Casino	Slot and roulette	<code>/minigames</code>	Appendix H
Tournaments	Waiting, play, results, winner side bets	<code>/tournaments/..</code> .	Appendix F
Social	Friends, invites, loans	<code>/friends,</code> lobby flows	Appendix I
Progression	XP, badges, daily challenges, login streak, leaderboard	<code>/profile,</code> <code>/leaderboard</code>	Appendix J
Account	Auth, 2FA, avatars, profile edit	<code>/auth,</code> <code>/edit-profile</code>	Appendix K
Admin	Operator web console (JWT admin role)	<code>/admin/console</code>	Appendix L
Ops	Metrics, health, CI, packaging	N/A (infra)	Appendix L
Other UX	Gift codes, free recharge, post-game rating modal	<code>client/src/</code> <code>components/</code> <code>Layout.tsx,</code> lobby	Appendix L

Domain	User-visible capability	Primary UI entry	Deep dive
Feedback	User feedback and player reports	Lobby/settings flows	Appendix L
Tournament UX	ZipRush mini-game while waiting	<code>client/src/features/tournament/components/ZipRushMiniGame.tsx</code>	Appendix F
Demo	Example and deal teaching routes	<code>/game-example, /game-deal</code>	Appendix B
A11y	Accessibility, table theming, ambient audio	<code>client/src/contexts/AccessibilityContext.tsx, TableThemeContext.tsx, MusicContext.tsx</code>	Appendix A

8 System Architecture

8.1 Monorepo layout

The repository splits into `client/` (Vite SPA), `server/` (Express + Socket.IO gateway), and `database/` (local Postgres). Client bootstraps providers in `client/src/main.tsx` and routes in `client/src/App.tsx`. Server composes middlewares and REST mounts in `server/src/index.ts`, then attaches `server/src/sockets/game.gateway.ts` to the HTTP server.

8.2 Real-time communication

Socket.IO uses path `/socket.io`. Authentication mirrors REST JWT validation (`server/src/middleware/socketAuth.middleware.ts`). Redis adapter enables horizontal scale-out when Redis is available.

8.3 Persistence and integrity

Prisma models capture users, games, tournaments, loans, hidden bet tickets, wallet ledger rows, and more (see `server/prisma/schema.prisma`). Critical monetary flows favor server-side orchestration and short transactions.

8.4 Data model overview (Prisma)

The persistence layer is relational PostgreSQL accessed through Prisma. Representative entities include: `User` (credentials, profile, chips, XP fields, avatar storage keys), `Game/seat` models for poker tables, `BlackjackRoom` and related seat/invitation models for multiplayer blackjack, tournament tables for bracket state and winner-bet tickets, `Friendship` and messaging hooks, loan and ledger tables for social/casino flows, and audit-friendly structures for wallet history. Full cardinality and migration history are only authoritative in `server/prisma/schema.prisma` and the migration folder.

8.5 Diagram

Figure 1 summarizes the request path. For a publication-quality diagram, export a Mermaid figure from `docs/ARCHITECTURE.md` to PNG and include it with `\includegraphics`.

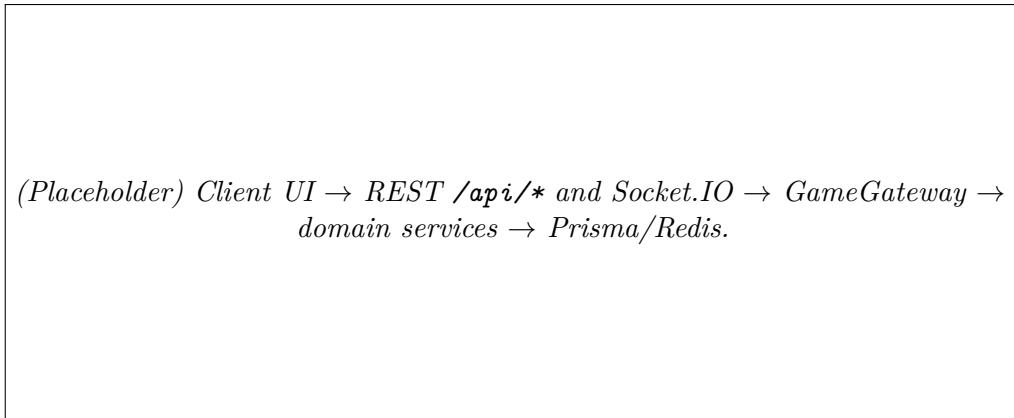


Figure 1: High-level runtime flow (placeholder box; replace with exported diagram if desired).

9 Deployment and operational runbook

9.1 Local development stack

The root script `npm run dev` (see `package.json`) launches PostgreSQL through `database/docker-compose.yml`, a lightweight helper script (`scripts/dev-ai.js`), the Node/Express server (`server/`), and the Vite client (`client/`) under concurrently. Redis should be available for rate limiting, action logging, and the Socket.IO Redis adapter when exercising multi-instance paths (`server/src/config/socketIoRedis.ts`). First-time setup, tester onboarding, and troubleshooting are documented in `README.md`, `Docs/SETUP_TESTEUR.md`, and `Docs/QUICK_TEST_GUIDE.md`.

9.2 Production and desktop delivery

VM-oriented deployment (Node, PostgreSQL, Redis, PM2), environment files, and Electron auto-update mechanics are described in `Docs/DEPLOY.md`. Mobile/desktop packaging scripts are exposed from `client/package.json` (Capacitor and Electron targets). Reverse-proxy samples appear under `nginx/` when applicable.

10 Security, Testing and Stability

10.1 Authentication and abuse controls

JWT on HTTP and sockets; optional TOTP 2FA (`server/src/routes/twofa.routes.ts`); Helmet and CORS allowlists (`server/src/index.ts`); segmented rate limits for bot, slot, roulette, blackjack, hidden-bets, and multi-blackjack routes; anti-cheat middleware and gateway-side monitors (`server/src/services/antiCheat.service.ts`, `server/src/sockets/game.gateway.ts`); idempotency middleware for sensitive mutations (`server/src/middleware/idempotency.middleware.ts`).

10.2 Testing and debugging

Broad automated coverage on server and client; Playwright smoke tests. Additional manual scenarios are documented under `Docs/POKER_SCENARIOS.md` and related files.

10.3 Automated test inventory and CI (reference snapshot)

Table 2 aggregates counts recorded in `Docs/RAPPORT_TESTS.md` (March 2026 snapshot). Teams should re-run Jest and Vitest after major refactors and update the markdown source of truth.

Layer	Runner	Suite files	Tests
Backend (<code>server/</code>)	Jest	12	257
Frontend (<code>client/</code>)	Vitest	6	20
Total automated	—	18	277

Table 2: Automated test inventory (representative snapshot; see `Docs/RAPPORT_TESTS.md`).

GitLab CI configuration at `.gitlab-ci.yml` encodes build and test gates for merge requests. A consolidated validation narrative for bot flows and UI smoke checks appears in `Docs/rapport-validation.md`.

10.4 Performance and stability

Redis-backed Socket.IO adapter, structured logging, Prometheus metrics endpoint, readiness endpoints, graceful shutdown hooks (`server/src/index.ts`).

11 Technical Challenges

Concurrency around shared tables: poker and blackjack runtimes use locking strategies (see `server/src/poker/services/pokerTableLock.service.ts` and blackjack lock services) to prevent split-brain chip mutations.

Recovery after process restarts: blackjack and tournament subsystems implement recovery services invoked at boot (`server/src/index.ts`).

Hidden bet settlement coupling: market resolution must align with poker street transitions; implemented in hidden-bet resolver modules (Appendix E).

Internationalization without clobbering user preference: guest screens (StartScreen, Auth, global loader) force English using `react-i18next` with a fixed `lng: 'en'` override so the global language stored for authenticated use is unchanged (`client/src/pages/Auth.tsx`, `client/src/contexts/LoaderContext.tsx`).

12 Ethics, privacy, and academic integrity

12.1 Virtual economy and regulatory framing

Quantum Bluff uses **in-game virtual chips** for learning and demonstration. It is not a licensed real-money gambling operator; features that resemble casino products are implemented in an **academic engineering** context and should be evaluated as software artefacts, not as a commercial gaming offer.

12.2 Personal data

Accounts store identifiers needed for authentication and gameplay (email, username, hashed password, optional TOTP secrets, profile metadata). Operators deploying a public instance must align hosting, retention, and breach notification with applicable law (including GDPR for EU users). This report does not replace a data protection impact assessment.

12.3 Integrity of authorship and AI assistance

Section 6 states how large language models were used as drafting assistants. All merged code remained subject to human review, tests, and repository governance. Each author is expected to sign off on their contribution statement in Section 13.

13 Individual Contributions

13.1 Omar – project management and architecture alignment

Organizational narrative: Section 4. Beyond ceremonies and backlog arbitration, Omar coordinated **cross-cutting technical commitments**: early adoption of Redis for Socket.IO horizontal scale-out and rate limiting, alignment between Prisma schema changes and gameplay features, and risk reviews before widening scope (tournaments, casino mini-games, loans). Interface with the appointed technical lead focused on sequencing risky merges (socket gateway, recovery jobs) away from demo deadlines.

13.2 Other contributors

The repository does not contain signed-off, per-person effort logs, story point assignments, or validated textual attributions beyond Omar’s narrative above. To satisfy examination requirements, the table below must be **completed and countersigned by the team** (replace placeholder cells).

Member	Primary areas (self-declared)	Verifiable artefacts (commits, MRs, modules)
Linda	<i>[to complete]</i>	<i>[to complete]</i>
Massi	<i>[to complete]</i>	<i>[to complete]</i>
Abdoulaye	<i>[to complete]</i>	<i>[to complete]</i>
Soheil	<i>[to complete]</i>	<i>[to complete]</i>
Mohamed	<i>[to complete]</i>	<i>[to complete]</i>
Azra	<i>[to complete]</i>	<i>[to complete]</i>
Yigit	<i>[to complete]</i>	<i>[to complete]</i>

Table 3: Individual contributions (template — team must fill before submission).

14 Limitations and Future Work

14.1 Current limitations

- **Documentation volume vs. examiners' time:** internal generated reports (e.g., `Docs/Rapport_final/RAPPORT_TECHNIQUE_COMPLET.md`) are useful as indexes but are not all human-curated line-by-line.
- **Dual loader providers:** `client/src/main.tsx` and `client/src/App.tsx` both wrap `LoaderProvider`; harmless but confusing—should be deduplicated.
- **Packaging surface area:** Electron/Capacitor increase release engineering work (signing, store policies) beyond the core web path.

14.2 Future improvements

Hardening continuous deployment secrets, expanding observability dashboards, deeper bot modeling (still code-first), and richer tournament analytics.

15 Conclusion

Quantum Bluff exceeded its initial poker-only charter by integrating multiple casino modes, tournaments, social lending, progression systems, and operational hardening while keeping a monorepo with clear separation between UI, API, and authoritative game logic. The project demonstrates that a student team can approach industry-like constraints—real-time state, monetary integrity, security, and observability—when management explicitly trades early waterfall rigidity for controlled agile iteration.

Lessons retained for similar efforts: (1) anchor scope changes to **recoverable increments** (feature flags, migrations, recovery jobs); (2) treat generated documentation as **hypotheses** until cross-read against code; (3) keep **security and idempotency** on the critical path rather than as end-of-project polish.

Acknowledgements

The team thanks the University of Strasbourg Computer Science faculty for supervising the L3 capstone framework, the open-source maintainers of the libraries listed in Section 5, and peers who provided feedback during playtests. Replace or extend this paragraph with institution-specific acknowledgements as required by the course booklet.

A Repository map and routing

A.1 Programmed routing and shell

Client routes are declared in `client/src/App.tsx`: public splash and auth (`/`, `/auth`, `/auth/admin`); authenticated gameplay and hub routes under `ProtectedRoute` with `Layout` (lobby, poker, blackjack, tournaments, mini-games, profile, friends, admin console). Admin routes use `AdminProtectedRoute`. Global providers stack in `client/src/main.tsx` (error boundary, loader, Redux, toasts, user, socket, Quantum HUD, hidden bets, audio) with an additional nested loader/invitation shell in `App.tsx`. Server REST mounts, rate-limit tiers, and health/metrics endpoints are wired in `server/src/index.ts`. A narrative overview also exists in `docs/ARCHITECTURE.md`.

B Poker engine and cash game

B.1 What was programmed

Authoritative table logic lives in `server/src/logic/GameTable.ts` (seats, streets, pot and side pots, action legality, showdown ordering) and `server/src/logic/CashGameController.ts` (chip debits/credits, transitions between phases). Hand evaluation and category ordering use `server/src/logic/Evaluator.ts`. Real-time delivery and timers (e.g. `TURN_TIMER` with 20s auto-fold/check) are implemented in `server/src/sockets/game.gateway.ts` together with anti-cheat hooks. The client implements presentation and input in `client/src/pages/Game.tsx` and reusable components under `client/src/components/` (`PokerTable`, `ActionButtons`, `CommunityCards`, `chat`, `HUD` hooks). **All wagering state changes are validated and applied on the server**; the client never “decides” the winner.

B.2 Rules enforced

Texas Hold'em flow from blinds through preflop, flop, turn, river, and showdown; fold, check, call, raise semantics; minimum raise increments; side pots when players are all-in for different amounts. Waiting-room creation and join flows are handled by dedicated routes (`server/src/routes/waitingRoom.routes.ts`, `server/src/routes/game.api.routes.ts`) coordinated with the gateway.

B.3 Further reading

Structured logic notes: `Docs/detail_logique/backend/01_poker_moteur.md`, `Docs/detail_logique/backend/02_cashgame_controller.md`.

C Bot AI and practice integration

C.1 Programmed pipeline

POST `/api/bot/action` (`server/src/routes/bot.routes.ts`) validates difficulty and cards, then calls `server/src/logic/botAI.ts`. Practice tables chain bot turns through `server/src/poker/services/practiceBotTurns.service.ts`; HTTP orchestration and optional expert delegation sit in `server/src/services/botAi.service.ts`. Four difficulties are supported: `easy`, `medium`, `hard`, `expert`.

C.2 Strength signal and decisions

Let H be the scalar hand score returned by `getHandValue` for the bot’s visible cards (two to seven cards). The code normalizes

$$\sigma = \min\left(1, \max(0, H/S_{\max})\right), \quad S_{\max} = 9 \cdot 15^5 + 14 \cdot 15^4,$$

matching the encoding used by `server/src/logic/Evaluator.ts`. Each difficulty combines σ with pot odds (`callAmount`, `potSize`), heads-up vs. multiway flags, and pseudo-random draws to return `FOLD`, `CALL`, `CHECK`, or `RAISE` with an optional raise amount (chips rounded via `intChips`). The easy tier, for example, injects naive bluffs on a free street with fixed probability and biases calls when facing small bets with medium strength.

C.3 Further reading

Design notes: `Docs/IA_EXPERT_BOT.md`.

D Probabilities, Quantum HUD, and Monte Carlo helper

D.1 Division of responsibilities

The **Quantum HUD** (`client/src/components/QuantumHUD.tsx`, context `client/src/contexts/QuantumHUDContext.tsx`) displays *estimates* for the authenticated player. These are computed client-side for responsiveness and are **not** used to settle real-money outcomes (the product uses virtual chips). Server-side hidden-bet pricing uses separate tabular services (Appendix E).

D.2 Monte Carlo equity estimator

Implementation file: `client/src/utils/pokerMonteCarloEquity.ts`. Let N be the number of iterations (default $N = 2000$, constant `MONTE_CARLO_DEFAULT_ITERATIONS`). For each iteration the code:

1. Builds the remaining deck excluding known hero and community cards (`buildDeckExcluding`).
2. Draws, without replacement, the missing board cards and $2 \times n_{\text{opp}}$ opponent hole cards using a partial Fisher–Yates swap sampler (`drawWithoutReplacement`).
3. Evaluates the hero and each opponent seven-card hand with `evaluateSevenEval` (same taxonomy as the server evaluator).
4. Splits the simulated pot equally among all players that achieve the best score.

If $S_t \in \{0, 1/k\}$ is the hero’s share on iteration t when k players tie for the best score, the reported equity is

$$\hat{e} = \frac{1}{N} \sum_{t=1}^N S_t.$$

A histogram of category indices (0–9) is accumulated in parallel for UX badges.

D.3 Monte Carlo error budget (HUD analogue to RTP)

The HUD does **not** implement a house-banked game: there is no monetary **RTP** in the casino sense. The quantity $\hat{e}$ is an **equity** estimate (expected share of the pot at showdown under uniform completion of unknown cards). If e denotes the true value of that expectation under the model, then $\mathbb{E}[\hat{e}] = e$ (each draw is independent and fair with respect to the remaining deck). Because $0 \leq S_t \leq 1$,

$$\text{Var}(S_t) \leq \frac{1}{4}, \quad \text{Var}(\hat{e}) = \frac{\text{Var}(S_t)}{N}, \quad \text{SE}(\hat{e}) \leq \frac{1}{2\sqrt{N}}.$$

With the shipped default $N = 2000$, $\text{SE}(\hat{e}) \leq 1/(2\sqrt{2000}) \approx 0.01118$ (about 1.1 equity points at most in the worst-case variance bound).

D.4 Quantum Combi—exact enumeration formula

The promotional “Quantum Combi” category (documented for players in `Docs/final_report/QUANTUM_COMBI.adoc`) requires ranks $\{3, 5, 6, 7, 10\}$ each to appear **at least once** among the seven cards (two private + five community). All $\binom{52}{7}$ seven-card subsets are equally likely:

$$\binom{52}{7} = 133784560.$$

For each rank r in that set, let A_r be the event “no card of rank r among the seven”; then $|A_r| = \binom{48}{7}$. By the principle of inclusion–exclusion over five ranks,

$$\left| \bigcup_r A_r \right| = \sum_{k=1}^5 (-1)^{k+1} \binom{5}{k} \binom{52-4k}{7} = 133002736.$$

Hence the favourable count is $133784560 - 133002736 = 781824$ and

$$P(\text{Quantum Combi}) = \frac{781824}{133784560} \approx 0.00584 \approx 0.584\%.$$

D.5 Quantum Combi in the hand-ranking ladder (server and HUD)

Both `server/src/logic/Evaluator.ts` and `client/src/Utils/pokerHandCategory.ts` use the same **best-of-seven** cascade. Categories are integers 0 (high card) through 9 (straight flush); a larger packed score beats a smaller one. After ruling out straight flush (9), four-of-a-kind (8), and full house (7), the code tests **Quantum Combi before flush or straight**. Quantum applies iff every rank in $\{3, 5, 6, 7, 10\}$ appears at least once in the seven rank values (Ace is coded as 14). Let V be the multiset of those seven values; subtract one occurrence of each mandatory rank; sort the remaining values descending and take the top two as kickers (κ_1, κ_2) . The score is

$$\text{packScore}(6, \kappa_1, \kappa_2) = 6 \cdot 15^5 + \kappa_1 \cdot 15^4 + \kappa_2 \cdot 15^3,$$

padding unused kicker slots with zeros in the implementation. Only if Quantum fails does the evaluator continue to flush (5), straight (4), three-of-a-kind, two pair, one pair, high card. Thus Quantum is positioned **strictly between full house and flush**, as a design choice documented in `Docs/final_report/QUANTUM_COMBI.adoc`.

E Hidden betting system

E.1 Programmed architecture

Hidden side markets are implemented under `server/src/poker/hiddenBets/`: typed markets and snapshots (`markets.ts`, `types.ts`), pricing tables (`pricing/pricingTables.ts`, `pricingTables.v1.ts`), live heuristics (`pricing/livePricing.ts`), quote and placement services, ledger context, and street-aware resolvers (`resolver/hiddenBetResolver.ts`). HTTP routes are mounted in `server/src/routes/hiddenBets.routes.ts`.

E.2 Odds and tables (example)

Live markets use tabulated decimal odds keyed by table parameters. For instance, `oddsPlayerWinsCurrentHand` maps the number of active players $n \in [2, 9]$ to a fixed grid (e.g. $n = 2 \rightarrow 1.9$, $n = 5 \rightarrow 3.0$, $n = 9 \rightarrow 4.6$); companion functions cover showdown vs. fold endings. These values are **heuristic** prices distanced from the pre-hand model (see code comments `hidden-bets-live-v1`), not Monte Carlo outputs from the HUD.

E.3 Further reading

`Docs/intern/HIDDEN_BETS_SPEC.md`, `Docs/detail_logique/backend/07_hidden_bets_moteur.md`.

F Tournaments and winner side bets

F.1 Programmed modules

Tournament lifecycle (registration, waiting room, table assignment, blind escalation, elimination, results) is implemented under `server/src/tournament/` with recovery jobs invoked at server boot (`server/src/index.ts`). Winner-side markets reuse wallet patterns in `server/src/tournament/winnerBets/tournamentWinnerBet.service.ts`. The client feature pack lives in `client/src/features/tournament/` (waiting room, in-tournament table, results, winner-bet panel). A lightweight `ZipRushMiniGame` retains players during queue phases.

F.2 Further reading

`Docs/intern/TOURNAMENT_FLOW_REAL_VS_EXPECTED.md`.

G Blackjack solo and multiplayer

G.1 Solo blackjack engine

Authoritative rules are implemented in `server/src/logic/blackjack.ts`. The server uses a six-deck shoe (`DECKS = 6`), shuffled in-place with Fisher–Yates (`shuffleShoe`). Card values follow standard rules: faces and tens count as 10, aces count as 11 when possible without exceeding 21, otherwise as 1. The dealer draws until the hard/soft total is at least 17; because aces are valued optimistically, the dealer **stands on soft 17** (e.g. A+6). Natural blackjack pays a 3:2 bonus on the stake (implemented as integer chip rounding per `intChips`); other wins pay 1:1; pushes return the stake. Double down is permitted only with exactly two player cards and issues one extra card.

Solo routes: `server/src/routes/blackjack.routes.ts`. Multiplayer rooms and seats: `server/src/routes/blackjackMulti.routes.ts`, controllers under `server/src/logic/BlackjackTableC` `ts`, in-memory runtime maps with recovery under `server/src/blackjack/`. UI: `client/src/pages/Blackjack.tsx`, lobby/table flows, components in `client/src/components/blackjack/`.

G.2 Further reading

`Docs/detail_logique/backend/03_blackjack_solo_moteur.md`, `Docs/detail_logique/backend/04_blackjack_multi_moteur.md`.

H Slot machine, roulette, and casino wallet

H.1 Three-reel slot model

The server module `server/src/logic/slotMachine.ts` draws three independent reels. Symbol weights are

$$w_{\text{cherry}} = 32, w_{\text{lemon}} = 24, w_{\text{bell}} = 18, w_{\text{seven}} = 14, w_{\text{diamond}} = 8, \quad W = \sum_i w_i = 96.$$

One reel satisfies $\Pr[R=i] = w_i/W$. Triples satisfy $\Pr(R_1=R_2=R_3=s) = (w_s/W)^3$. Triple s pays total return multiplier $m_s \in \{5, 8, 10, 15, 20\}$ on the stake; an **exact pair** (two equal symbols, third different) pays multiplier 1 (stake returned, zero net profit); otherwise the payout multiplier is 0. RNG is injectable (`RandomIntFn`, default `crypto.randomInt` via `server/src/rng/rng.service.ts`).

H.2 Complete slot RTP (closed form, current constants)

Let $X = \text{winAmount}/\text{bet}$ be the **gross return multiplier** (total coins returned by the machine, including the stake on a push). Write R_1, R_2, R_3 for the three reel symbols.

Triple (three of a kind).

$$\Pr(\text{triple of } s) = \left(\frac{w_s}{W}\right)^3, \quad \Pr(\text{any triple}) = \sum_s \left(\frac{w_s}{W}\right)^3 = \frac{\sum_s w_s^3}{W^3}.$$

Numerically $\sum_s w_s^3 = 32768 + 13824 + 5832 + 2744 + 512 = 55680$ and $W^3 = 96^3 = 884736$, hence

$$\Pr(\text{any triple}) = \frac{55680}{884736} \approx 0.0629 \text{ (6.29\%)}. \quad \text{}$$

Exact pair (not a triple). For a doubled symbol A ,

$$\Pr(\text{pair with } A \text{ doubled}) = \frac{3 w_A^2 (W - w_A)}{W^3}, \quad \Pr(\text{any pair}) = \sum_A \frac{3 w_A^2 (W - w_A)}{W^3}.$$

Numerically $\sum_A 3 w_A^2 (W - w_A) = 461952$, so $\Pr(\text{pair}) = 461952/884736 \approx 0.5221$ (52.21%).

Lose.

$$\Pr(\text{lose}) = 1 - \Pr(\text{triple}) - \Pr(\text{pair}) \approx 0.4149 \text{ (41.49\%)}. \quad \text{}$$

Expected multiplier and RTP. Because outcomes partition $\{\text{lose, pair, triples}\}$,

$$\mathbb{E}[X] = \sum_s \left(\frac{w_s}{W}\right)^3 m_s + \Pr(\text{pair}) \cdot 1 = \frac{\sum_s m_s w_s^3 + 461952}{884736}.$$

Here $\sum_s m_s w_s^3 = 5 \cdot 32768 + 8 \cdot 13824 + 10 \cdot 5832 + 15 \cdot 2744 + 20 \cdot 512 = 384152$, so

$$\mathbb{E}[X] = \frac{384152 + 461952}{884736} = \frac{846104}{884736} \approx 0.9563.$$

The **return-to-player (RTP)** expressed as a percentage of the stake is $100 \mathbb{E}[X] \approx 95.63\%$. The **house edge** on the stake is $1 - \mathbb{E}[X] \approx 4.37\%$. The expected **net** gain per spin is $\mathbb{E}[G] = \mathbb{E}[\text{winAmount} - \text{bet}] = \text{bet} (\mathbb{E}[X] - 1) \approx -0.0437 \times \text{bet}$.

Variance (optional diagnostic). With the same X , the AsciiDoc reference reports $\text{Var}(X) \approx 3.12$ (standard deviation ≈ 1.77); hence $\text{Var}(G) = \text{bet}^2 \text{Var}(X)$ for a flat bet. See `Docs/SLOT_MACHINE.adoc` for the full derivation.

H.3 European roulette model

The server module `server/src/logic/roulette.ts` draws a single winning integer uniformly from $\{0, \dots, 36\}$ (`crypto.randomInt` through the same RNG service). Payouts follow the European **total-return** multipliers on the stake for winning pockets (straight 36, split 18, street 12, corner 9, six-line 6, dozen/column 3, even-money 2). Zero causes outside bets to lose unless covered.

H.4 Complete roulette RTP (all proportional inside bets)

Consider any bet line that covers exactly k distinct winning pockets ($1 \leq k \leq 18$ on the proportional ladder used in code) with European gross multiplier $m = 36/k$ applied to the stake when one of those pockets hits. Because each pocket has probability $1/37$,

$$\mathbb{E}\left[\frac{\text{payout}}{\text{bet}}\right] = \frac{k}{37} \cdot m = \frac{k}{37} \cdot \frac{36}{k} = \frac{36}{37} \approx 0.97297.$$

Hence every such line has the same **RTP** $= 36/37 \approx 97.297\%$ of the stake returned as gross payout in expectation, and the same **house edge** $= 1/37 \approx 2.703\%$ on the stake. The expected **net** gain per unit staked on a single line is

$$\frac{k}{37} \cdot m - 1 = -\frac{1}{37}.$$

When several independent bet lines are placed on one spin, expectation adds across lines; the server enforces caps (`ROULETTE_MAX_BET_CAP`, `ROULETTE_MAX_TOTAL_STAKE`, `ROULETTE_MAX_BETS_PER_SPIN`) before settlement. Wheel animation order is `EUROPEAN_WHEEL_ORDER`. Full payout tables: `Docs/final_report/ROULETTE.adoc`.

H.5 Wallet and ledger

Casino flows integrate with wallet/ledger helpers under `server/src/casino/` and `server/src/wallet/`. Additional narrative: `Docs/detail_logique/backend/05_roulette_moteur.md`, `Docs/detail_logique/backend/06_slot_moteur.md`, `Docs/final_report/ROULETTE.adoc`.

I Social graph, invitations, and peer loans

I.1 Programmed surfaces

Friends UI: `client/src/pages/Friends.tsx`. REST: `server/src/routes/friends.routes.ts`; poker invitations `server/src/routes/invitation.routes.ts` (also under `/api/invitations`); blackjack room invitations share patterns with the lobby. Peer loans: `server/src/routes/friendLoan.routes.ts`, business logic in `server/src/services/friendLoan.service.ts`, real-time updates via `server/src/services/friendLoan.emit.ts` and socket listeners on the client.

I.2 Further reading

`Docs/detail_logique/backend/09_friend_loans_moteur.md`.

J Progression, challenges, and leaderboards

J.1 Programmed logic

Shared rules in `server/src/logic/gamification.ts` (XP curves, badge triggers). Daily challenges and login streak are isolated modules (`server/src/dailyChallenges/`, `server/src/dailyLogin/`) with dedicated routers. Public leaderboards: `server/src/routes/leaderboard.routes.ts`; client pages under `/leaderboard` and profile summaries.

J.2 Further reading

`Docs/detail_logique/backend/10_gamification_moteur.md`.

K Authentication, profiles, and i18n

K.1 Programmed security and profiles

JWT issuance and verification on `server/src/routes/auth.routes.ts`; password hashing (bcrypt) and validation schemas (Zod). Optional TOTP 2FA: `server/src/routes/twofa.routes.ts`. Client auth flows: `client/src/pages/Auth.tsx`. Profile and avatar uploads: `client/src/pages/EditProfile.tsx` with binary fields or URLs on the Prisma `User` model. Internationalization bundles live in `client/src/i18n/locales/` (five languages); guest-only screens force English overlays where documented in Section 11.

L Operations, Redis, CI, and packaging

L.1 Programmed operations

Redis clients for Socket.IO horizontal scale-out: `server/src/config/socketIoRedis.ts`. Rate limiting and action logs share Redis primitives configured in `server/src/index.ts`. Observability helpers: `server/src/observability/`. CI pipeline: `.gitlab-ci.yml`. Desktop/mobile packaging scripts in `client/package.json` (Electron, Capacitor). Admin web console: `client/src/pages/AdminConsole.tsx` with `server/src/routes/adminConsole.routes.ts`. Dangerous admin/dev routes are compiled out or gated when `env.isProduction` is true.

M External references and compendium

- Initial formal requirements PDF: `Docs/Cahier_de_charge_6A_PI.pdf` (not embedded here).
- Auto-generated wide-coverage technical compendium: `Docs/Rapport_final/RAPPORT_TECHNIQUE_COMPLET.md` (use as index; verify claims against code).
- Security dossiers: `Docs/Architecture/SECURITE_A_TO_Z.md`, `Docs/security/`.
- HTTP and socket maps: `Docs/detail_logique/network/01_http_api_map.md`, `Docs/detail_logique/network/02_socket_events_map.md`.
- Combination analysis report: `Docs/final_report/QUANTUM_COMBI.adoc`.
- Slot RTP and outcome algebra (AsciiDoc): `Docs/SLOT_MACHINE.adoc`.
- Roulette payouts and expectation: `Docs/final_report/ROULETTE.adoc`.

References

- [1] Meta. *React documentation* (React 19). <https://react.dev/> (accessed May 2026).
- [2] Microsoft. *TypeScript handbook*. <https://www.typescriptlang.org/docs/> (accessed May 2026).
- [3] Vite contributors. *Vite guide*. <https://vite.dev/guide/> (accessed May 2026).
- [4] OpenJS Foundation. *Node.js runtime documentation*. <https://nodejs.org/docs/latest/api/> (accessed May 2026).
- [5] OpenJS Foundation. *Express web framework*. <https://expressjs.com/> (accessed May 2026).
- [6] Socket.IO project. *Socket.IO server and client (v4)*. <https://socket.io/docs/v4/> (accessed May 2026).
- [7] Prisma Data Inc. *Prisma ORM documentation*. <https://www.prisma.io/docs> (accessed May 2026).
- [8] PostgreSQL Global Development Group. *PostgreSQL documentation*. <https://www.postgresql.org/docs/> (accessed May 2026).
- [9] Redis Ltd. *Redis documentation*. <https://redis.io/docs/> (accessed May 2026).
- [10] Redux team. *Redux Toolkit and RTK Query*. <https://redux-toolkit.js.org/> (accessed May 2026).
- [11] i18next authors. *i18next documentation*. <https://www.i18next.com/> (accessed May 2026).
- [12] Zod maintainers. *Zod schema validation*. <https://zod.dev/> (accessed May 2026).
- [13] OpenJS Foundation. *Jest testing framework*. <https://jestjs.io/> (accessed May 2026).
- [14] Vitest contributors. *Vitest*. <https://vitest.dev/> (accessed May 2026).
- [15] Microsoft. *Playwright end-to-end testing*. <https://playwright.dev/> (accessed May 2026).
- [16] Tailwind Labs. *Tailwind CSS documentation*. <https://tailwindcss.com/docs> (accessed May 2026).
- [17] Schwaber, K., Sutherland, J. *The Scrum Guide* (2020). <https://scrumguides.org/scrum-guide.html> (accessed May 2026).
- [18] GitLab Inc. *GitLab CI/CD documentation*. <https://docs.gitlab.com/ee/ci/> (accessed May 2026).
- [19] Université de Strasbourg. *Faculté des mathématiques et informatique*. <https://mathinfo.unistra.fr/> (accessed May 2026).